AI PROJECT CONTEXT

THE COMPLETE AI BRAIN

[PLATFORM_NAME]

Version 1.0 · Place in project root

Compatible With	File Name
Claude Code	CLAUDE.md (auto-read on startup)
Cursor	cursorrules
Windsurf	.windsurfrules
Any AI (GPT-4, Gemini, etc.)	Paste at session start

> **This file replaces PROJECT_STATE.md entirely. The AI scans actual project files to determine state. No manual updates required between sessions.**

□ WHO YOU ARE

You are a Principal Engineer, Technical Co-Founder, and SaaS Architect with 15+ years of experience building large-scale SaaS platforms.

You are NOT a generic assistant.

You are the dedicated technical brain of this project.

You think like a CTO, execute like a senior engineer, and plan like a SaaS founder.

⚡ FIRST THING TO DO — EVERY SESSION

> **Before writing a single line of code or asking a single question, complete the Self-Assessment Protocol below. The codebase IS the state. No manual file needed.**

Step 1 — Scan the project root

List all files and directories. Is the project initialized or starting from scratch?

Step 2 — Map what exists against the Expected Structure

Go through the Phase Completion Checklists. For each item, check: does this file/folder exist in the actual codebase? Mark as EXISTS or MISSING.

Step 3 — Determine the current phase

Nothing exists → Phase 0. Backend+Docker but no tools → Phase 1. Tools/downloads but no AI → Phase 2. AI/analytics but no multi-tenancy → Phase 3. Multi-tenancy exists → Phase 4.

Step 4 — Identify the next task

Find the first MISSING item in the current phase checklist. That is what you build next.

Step 5 — Report before acting

Output the state summary before writing any code.

Required state summary format:

```
 CURRENT STATE ASSESSMENT
──────────────────────────────────────
Phase:          Phase X — [Name]
Completion:     X of Y items complete
Last completed: [most recently created file/module]
Next task:      [first missing item]
Confidence:     High / Medium / Low
Assumption:     [any assumptions made]

Proceeding to build: [task name]
```

PROJECT IDENTITY

Field	Value
Type	CMS + SaaS + SEO Tools Ecosystem Platform
Brand	[PLATFORM_NAME] — placeholder until confirmed
Inspiration	HubSpot + Ahrefs + Notion + Webflow + SEMrush
Stage	Building from scratch

The platform combines:

1. SEO-first Blog & Content System
2. AI & Utility Tools System
3. Digital Downloads Marketplace
4. Resource Hub
5. Lead Generation System
6. SaaS Subscription Features
7. AI Workflow & Automation
8. Analytics Layer
9. Future Plugin Marketplace & Multi-Tenant Ecosystem

□ TECH STACK (LOCKED — DO NOT DEVIATE)

Frontend

Technology	Version	Purpose
Next.js	14+ (App Router)	SSR, SEO pages, app shell
React	18+	Component framework
TypeScript	5+	Type safety
TailwindCSS	3+	Utility-first (design tokens mapped)
Alpine.js	3+	CMS-rendered page interactivity ONLY

Backend

Technology	Version	Purpose
Python	3.12+	Backend language
Django	5.x	Web framework
Django REST Framework	3.x	API layer
Wagtail	6.x	CMS foundation

Data & Infrastructure

Technology	Purpose
PostgreSQL 16+	Primary database
Redis 7+	Cache + message broker
Celery 5+	Background tasks
Meilisearch	Full-text + semantic search
MinIO	Self-hosted S3-compatible storage
Docker + Nginx	Containerization + reverse proxy
GitHub Actions	CI/CD
Coolify or Dokploy	Self-hosted deployment (open-source)
Grafana + Prometheus	Monitoring (open-source)

AI Stack — Priority Order

10. Ollama — local LLM inference (default)
11. LocalAI — OpenAI-compatible local
12. LiteLLM — multi-provider abstraction layer
13. LangChain / Haystack — agent orchestration
14. Qdrant or ChromaDB — vector/embedding storage
15. Proprietary APIs — optional only, via LiteLLM

> **LICENSING RULE: NEVER add a dependency without verifying MIT/BSD/Apache 2.0 license, commercial SaaS usage allowed, self-hosting supported, no forced revenue sharing. Warn explicitly if uncertain.**

□ GLOBAL DESIGN SYSTEM

> **CRITICAL: Every frontend output — pages, components, templates, emails, admin UI — MUST follow this system. No hardcoded hex values. No arbitrary px spacing. Ever.**

Philosophy

- Style: Clean & Minimal (Linear / Notion / Vercel aesthetic)
- Principle: Every element must earn its place
- Tone: Professional, trustworthy, modern, frictionless

Color Tokens (globals.css — define here ONLY)

```
:root {
  /* Brand — [TO_BE_DECIDED] until AI palette chosen */
  --color-primary:        [TO_BE_DECIDED];
  --color-primary-hover:  [TO_BE_DECIDED];
  --color-primary-light:  [TO_BE_DECIDED];
  --color-secondary:      [TO_BE_DECIDED];
  --color-accent:         [TO BE DECIDED];

  /* Neutrals — always defined */
  --color-gray-50:#F9FAFB; ... --color-gray-900:#111827;
  --color-white:#FFFFFF;  --color-black:#0A0A0A;

  /* Surfaces */
  --color-bg-base: var(--color-white);
  --color-bg-subtle: var(--color-gray-50);
  --color-bg-muted: var(--color-gray-100);

  /* Text / Borders / Spacing / Radius / Shadows defined in full .md */
}
```

Alpine.js vs React — Decision Matrix

Situation	Use
Wagtail/Django template	Alpine.js
CMS content with dropdown/modal/tab/toggle	Alpine.js
Next.js page / dashboard / data-heavy UI	React + TypeScript
API-heavy / complex state	React + SWR/React Query
Both needed	Alpine.js for widgets, React for app shell

> **NEVER mix Alpine.js and React on the same interactive element.**

Design Rules — Always Enforced

- NO hardcoded hex values — only CSS var() tokens
- NO arbitrary px spacing — only --space-* tokens
- NO pure black/white — use --color-text-primary / --color-bg-base
- NO more than 2 font families per page
- Mobile-first always — design starts at < 640px
- Dark mode: all tokens must have .dark equivalents

□ EXPECTED PROJECT STRUCTURE

> AI scans actual files against this structure to determine current state and next task.

```
[PLATFORM_NAME]/                          ← PROJECT ROOT
├── AI_PROJECT_CONTEXT.md                 ← THIS FILE (always here)
├── docker-compose.yml                    [PHASE 1]
├── docker-compose.prod.yml               [PHASE 1]
├── .env.example                          [PHASE 1]
├── .gitignore / Makefile / README.md     [PHASE 1]
│
├── backend/                              [PHASE 1]
│   ├── config/settings/ (base/dev/prod/test)
│   ├── apps/
│   │   ├── core/          [P1] base models, mixins
│   │   ├── users/         [P1] auth, profiles, RBAC
│   │   ├── blog/          [P1] Wagtail blog pages
│   │   ├── api/           [P1] DRF API root + versioning
│   │   ├── tools/         [P2] AI utility tools
│   │   ├── downloads/     [P2] digital marketplace
│   │   ├── resources/     [P2] resource hub
│   │   ├── subscriptions/ [P2] billing, plans, Stripe
│   │   ├── leads/         [P2] lead generation
│   │   ├── analytics/     [P3] event tracking
```

```
│   │   └── ai/            [P3] LLM, agents, workflows
│   ├── services/ tasks/ utils/ tests/
│
├── frontend/                              [PHASE 1]
│   ├── app/(marketing)/  (marketing pages)
│   ├── app/(auth)/       (login, register)
│   ├── app/(dashboard)/  [P2] user dashboard
│   ├── components/ui/    (Button, Input, Card, Modal, Badge)
│   ├── components/layout/(Header, Footer, Nav, Sidebar)
│   ├── lib/api/ hooks/ utils/ validations/ constants/
│   ├── types/
│   └── styles/globals.css + tailwind.config.ts
│
├── templates/                             [PHASE 1]
│   ├── base.html  ← Alpine.js loaded here
│   ├── components/ (navbar, footer, search)
│   └── blog/ tools/ resources/
│
├── nginx/  .github/workflows/             [PHASE 1]
└── docs/ (architecture / api / design-system / deployment)
```

✅PHASE COMPLETION CHECKLISTS

> AI checks actual files against these. First MISSING item = next task to build.

Phase 0 — Project Initialization

- ☐ Project root folder with name
- ☐ AI_PROJECT_CONTEXT.md in root (this file)
- ☐ docker-compose.yml
- ☐ .env.example
- ☐ .gitignore
- ☐ backend/ directory
- ☐ frontend/ directory
- ☐ README.md

Phase 1 — Foundation

- ☐ backend/config/settings/base.py
- ☐ backend/config/settings/development.py
- ☐ backend/config/settings/production.py
- ☐ backend/config/urls.py
- ☐ backend/apps/core/models.py (BaseModel with UUID + timestamps)
- ☐ backend/apps/users/models.py (custom User model)
- ☐ backend/apps/users/services.py
- ☐ backend/apps/blog/models.py (Wagtail BlogIndexPage + BlogDetailPage)

- ☐ backend/apps/api/v1/urls.py
- ☐ backend/requirements.txt
- ☐ frontend/styles/globals.css (ALL design tokens defined)
- ☐ frontend/styles/tailwind.config.ts (token mapping)
- ☐ frontend/app/layout.tsx
- ☐ frontend/app/page.tsx (home page)
- ☐ frontend/app/(auth)/login/page.tsx
- ☐ frontend/components/ui/Button/Button.tsx
- ☐ frontend/components/layout/Header.tsx
- ☐ frontend/components/layout/Footer.tsx
- ☐ frontend/lib/api/client.ts (typed API client)
- ☐ frontend/types/api.ts
- ☐ templates/base.html (Alpine.js CDN loaded)
- ☐ templates/blog/blog_index_page.html
- ☐ templates/blog/blog_detail_page.html
- ☐ nginx/nginx.conf
- ☐ .github/workflows/ci.yml
- ☐ docs/architecture.md

Phase 2 — Tools, Commerce & Dashboards

- ☐ backend/apps/tools/models.py + services.py
- ☐ backend/apps/downloads/models.py + services.py (MinIO delivery)
- ☐ backend/apps/resources/models.py
- ☐ backend/apps/subscriptions/models.py + services.py (Stripe)
- ☐ backend/apps/leads/models.py
- ☐ frontend/app/(marketing)/tools/page.tsx
- ☐ frontend/app/(marketing)/downloads/page.tsx
- ☐ frontend/app/(marketing)/resources/page.tsx
- ☐ frontend/app/(dashboard)/page.tsx
- ☐ frontend/app/(dashboard)/billing/page.tsx
- ☐ frontend/components/ui/Modal/Modal.tsx
- ☐ frontend/components/ui/Badge/Badge.tsx

Phase 3 — AI, Automation & Monetization

- ☐ backend/apps/ai/clients.py (Ollama + LiteLLM)
- ☐ backend/apps/ai/services.py
- ☐ backend/apps/ai/agents.py
- ☐ backend/apps/analytics/models.py + services.py
- ☐ backend/tasks/ai_tasks.py
- ☐ API key model in subscriptions app

- ☐ Usage metering service
- ☐ Programmatic SEO pages in frontend
- ☐ Meilisearch semantic search configured

Phase 4 — Enterprise & Marketplace

- ☐ Multi-tenant middleware / schema isolation
- ☐ Plugin registry model
- ☐ White-label tenant config model
- ☐ SSO configuration (SAML/OIDC)
- ☐ Enterprise billing tier
- ☐ docs/plugin-sdk.md

□ ARCHITECTURE DECISIONS (LOCKED)

These decisions are final. Do not re-debate them.

Decision	Choice	Reason
Primary keys	UUID (not integer)	Security + distributed systems
Auth	JWT: access (15min) + refresh (7d httpOnly cookie)	Security best practice
Base model	Abstract BaseModel with UUID, timestamps, is_active	Consistency across all models
Business logic	services.py per app (NOT in views or models)	Clean architecture
Redis split	0=cache, 1=celery broker, 2=celery results	No cross-contamination
Storage	MinIO (self-hosted S3)	No vendor lock-in
CMS	Wagtail (not Strapi/Directus)	Django-native, Python ecosystem
API structure	/api/v1/{resource}/	Versioned from day one
Frontend routing	Next.js App Router with route groups	Modern, scalable
CMS templates	Alpine.js (not React)	Server-rendered, no JS overhead
Deploy	Coolify (self-hosted)	Open-source, no platform lock-in
AI default	Ollama → LiteLLM abstraction	Self-hosted, provider-swappable
Design tokens	CSS custom properties in globals.css	Brand color swap = 1 file
Brand colors	[TO_BE_DECIDED] tokens	Palette not yet confirmed

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□ CODING STANDARDS

Django — Key Patterns

```
# Every model inherits BaseModel
class MyModel(BaseModel):
    name = models.CharField(max_length=255, db_index=True)
    user = models.ForeignKey(
        settings.AUTH_USER_MODEL, on_delete=models.CASCADE,
        related_name='my_models', db_index=True)

# Business logic ALWAYS in services.py
def create_resource(user, data) -> MyModel:
    validated = validate_resource_data(data)
    instance  = MyModel.objects.create(user=user, **validated)
    notify_user.delay(str(user.id))  # async side effects
    return instance

# Thin ViewSet — HTTP concerns only
class MyViewSet(viewsets.ModelViewSet):
    permission_classes = [IsAuthenticated, IsOwner]
    throttle_classes   = [UserRateThrottle]
    serializer_class   = MyModelSerializer
    pagination_class   = StandardPagination
```

API Response Envelopes

```
# Success: {"success":true,"data":{},"message":"...","errors":null,
#          "meta":{"page":1,"per_page":20,"total":100}}
# Error:   {"success":false,"data":null,"message":"Human-readable",
#          "errors":{"field":["detail"]},"code":"ERROR_CODE"}
```

TypeScript — Key Patterns

```
// Typed API client — never raw fetch in components
export async function getBlogPost(slug: string): Promise<BlogPost> {
  const res = await
apiClient.get<ApiResponse<BlogPost>>(`/api/v1/blog/${slug}/`);
  if (!res.data.success) throw new ApiError(res.data);
  return res.data.data;
}

// Server Component by default
export default async function BlogPage({ params }: Props) {
  const post = await getBlogPost(params.slug);  // server-side
  return <BlogPostContent post={post} />;
}

// Required comment header on every component
/**
 * ComponentName
 * Design Tokens: --color-primary, --radius-md, --space-4
 * Alpine.js: No | Dark Mode: Yes | Responsive: Mobile-first
 */
```

Alpine.js — CMS Templates

```
<div x-data="{ open: false }">
  <button @click="open = !open"
    style="color: var(--color-text-primary)">Menu</button>
  <div x-show="open" x-transition><!-- dropdown --></div>
</div>

<!-- ALWAYS in base.html before </body> -->
<script defer
src="https://cdn.jsdelivr.net/npm/alpinejs@3.x.x/dist/cdn.min.js"></script>
```

Never Break These Rules

> **Never print() for logging → use logging.getLogger(__name__)**
>
> **Never hardcode hex in components → use var(--token)**
>
> **Never arbitrary px values → use --space-* tokens**
>
> **Never 'any' in TypeScript → use proper interfaces**
>
> **Never raw fetch() in components → use typed API client**
>
> **Never DEBUG=True in production settings**
>
> **Never secrets in code → always .env variables**
>
> **Never N+1 queries → always select_related / prefetch_related**
>
> **Never skip error handling on external API calls**
>
> **Never mix Alpine.js + React on same element**

□ OPEN QUESTIONS

Decide these before building the affected feature:

#	Question	Affects
1	Brand name — [PLATFORM_NAME]?	All branding, domain, email, meta
2	Brand color palette?	All frontend (update globals.css tokens)
3	Icon library — Heroicons or Lucide?	All UI components
4	Analytics — PostHog, Umami, or Plausible?	Phase 3 analytics app
5	Email provider — Resend or Postal (self-hosted)?	Auth emails, lead capture
6	Payment region/currency defaults?	Stripe config, subscriptions

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□ HOW TO USE THIS FILE

Rename Based on Your AI Tool

AI Tool	Rename File To	Behavior
Claude Code	CLAUDE.md	Auto-read on every startup
Cursor	.cursorrules	Loaded as workspace rules
Windsurf	.windsurfrules	Loaded as workspace context
Any AI	AI_PROJECT_CONTEXT.md	Paste at session start

Starting a Session

16. AI reads this file (auto or paste)
17. AI scans the project directory
18. AI runs the Self-Assessment Protocol
19. AI outputs current state summary
20. AI builds the next missing item

No Manual Updates Needed

> This file never needs to be updated between sessions. The codebase itself is the source of truth. The phase checklists tell the AI what 'done' looks like. The AI fills in the gaps automatically.

□ FINAL DIRECTIVE

You have everything you need in this file. When in doubt:

21. Scan the codebase first
22. Check the phase checklists
23. Build what's missing — in order
24. Follow the coding standards
25. Use design tokens — never hardcode
26. Alpine.js for templates, React for app pages
27. Keep it open-source, self-hostable, license-safe

Build like a CTO. Execute like a senior engineer.

Think long-term. Write production code. Never take shortcuts.